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ORIGINAL ARTICLE

Body composition assessment in children with inflammatory bowel disease: A comparison of different methods

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Aim: To assess different techniques to measure body composition in paediatric patients with inflammatory bowel disease using dual energy X-ray absorptiometry as a reference method. We hypothesised that a three-compartment model may demonstrate superiority over other methods as skinfold thickness equations and bioelectrical impedance analysis.

Methods: Body composition was assessed using skinfold thickness equations, bioelectrical impedance analysis and the three-compartment model. Data obtained with these methods were compared to the results obtained by dual energy X-ray absorptiometry. Statistical analysis was performed using Spearman's correlation and Bland–Altman's limits of agreement method.

Results: Twenty-one paediatric patients with inflammatory bowel disease were included: 11 females and 10 males; mean age for the entire group: 14.3 years, range 12–16 years. In children with inflammatory bowel disease, skinfold thickness equations, bioelectrical impedance analysis and the three-compartment model showed reliable measurements with small differences in the percentage of total body fat and good limits of agreements.

Conclusion: The assessment of body composition using bioelectrical impedance analysis provides a valid and accurate method in children with inflammatory bowel disease as compared to dual energy X-ray absorptiometry. In the future, superiority of 3-compartment model in research and clinical settings of nutritional intervention and disease status in children with inflammatory bowel disease remains to be demonstrated.

Key words: bioelectrical impedance analysis; children; dual energy X-ray absorptiometry; inflammatory bowel disease; skinfold equation; three-compartment model.

What is already known on this topic

1 Assessment of nutritional status using body composition is mandatory in children with chronic inflammatory diseases.
2 A number of techniques for measurement of body composition in children with inflammatory bowel disease (IBD) are available. Accuracy and precision of different methods are heterogeneous.
3 Dual-energy X-ray absorptiometry (DEXA) and four-compartment model serve as gold standard in clinical settings and research studies for assessment of body composition respectively.

What this paper adds

1 Skinfold thickness (SF), bioelectrical impedance analysis (BIA) and three-compartment (3-C) model to measure body composition provide valid and accurate assessment of body composition in children with IBD as compared to DEXA.
2 BIA (limits of agreement −6.4 to 6.5) showed the strongest correlation and agreement in comparison to the SF equation (limits of agreement −6.1 to 10.7) and 3-C model (−4.5 to 8.6) estimating BF% by DEXA as a reference method.
3 Since the 3-C model has not demonstrated superiority over other methods, potential use of BIA in research and clinical settings of future nutritional intervention and disease status in children with IBD should be considered.

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Conflict of interest: Francois-Pierre Martin, Isabelle Breton and Jean-Philippe Godin are employees of Société des Produits Nestlé (Switzerland).

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Approximately, 25% of all patients with inflammatory bowel disease (IBD) are diagnosed during childhood or adolescence.[1] Growth failure and delayed puberty are common features of paediatric patients with IBD. Nutritional status is however quite difficult to assess and cannot be limited by simple measures of weight or body mass index (BMI). Body composition assessment, which provides a better indicator of health and nutritional status, describes the proportions of fat mass, fat-free mass (FFM) and bone mineral density of an individual. Alterations in body

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composition are expected in children with inflammatory conditions as demonstrated by a systematic review revealing different amounts of fat mass and FFM in children with Crohn's disease (CD) compared to healthy subjects.[2] Various methods based on compartment models accessing body composition are available. Anthropometry, such as measuring skinfold thickness (SF) is inexpensive, safe and easy to perform in children, although its accuracy remains controversial.[3] Dual-energy X-ray absorptiometry (DEXA) and bioelectrical impedance analysis (BIA) are widely used for estimation of body composition in clinical research and research settings.[4,5] Although limited to research studies, the most accurate *in vivo* technique available is the four-compartment model, which allows an individual evaluation of varying components such as FFM, body density (BD), body water and bone mineral content. This method is considered to represent the gold-standard in adult and paediatric populations.[6,7] An alternative method is the three-compartment model (3-C model) developed by Siri which measures BF% from estimates of BD obtained from SF and total body water (TBW) from BIA.[8] Although DEXA is considered as a reference method for body composition assessment, this technique remains expensive and inconsistently present in clinical and research settings. Looking for a method with similar precision and accuracy without using sophisticated instrument is of interest in children with chronic illness, or in longitudinal studies where metabolic and nutritional outcomes are crucial purposes. Therefore, the aim of the present study was to assess the agreement of BF% measurement obtained using SF equations, BIA and 3-C model versus DEXA technique in a population of children with IBD. We hypothesised that a 3-C model may demonstrate superiority over other methods as SF equations and BIA.

Methods

Subjects

The present study is a cross-sectional study about growth and energy expenditure in children with IBD. Patients with IBD were recruited from August 2011 to October 2012 from the Pediatric Gastroenterology Outpatient Clinic of the University Hospital of Lausanne, Switzerland. Eligible patients were aged between 10 and 18 years, with a diagnosis of CD or ulcerative colitis (UC), confirmed according to international criteria.[9] All participants were diagnosed prior to study begin and in clinical remission according to their respective activity indexes.

Ethics

The study was approved by the Human Ethics Committee of the Canton de Vaud, Switzerland (protocol 69/10). Informed written consent was obtained from patients and their parents.

Assessment of body composition

Body composition measurements were performed at the same day at baseline, 6 and 12 months.

Anthropometrics

Weight was measured to the nearest 0.1 kg using a calibrated digital scale (Seca, Hamburg, Germany) in patients wearing underwear. Height was measured to the nearest 0.1 cm using a stadiometer (Holtain, Crosswell, UK). BMI was calculated as weight (kg)/height (m)^2. *Z*-scores for weight, height and BMI were obtained using EPI-Info 2002 software (Epidemiological software package for disease control and prevention, Atlanta, GA, USA). Z-scores between −2 and +2 were considered normal.

Skinfold thickness measurements

SFs were measured by the same trained examiner (TC) using a Harpenden Calliper (Chasmors Ltd., London, UK), according to the protocol of Lohman.[10] Measurements were performed based on the recommendation of Durnin and Womersley[11] at the following sites: (i) biceps (at the level of the mid-point between the lateral edge of the acromion process and the lateral border of the radius bone, on the mid-line of the anterior surface of the arm); (ii) triceps (at the level of the mid-point between the acromion process and the lateral border of the radius bone, on the mid-line of the posterior surface of the arm); (iii) subscapular (measured 1 cm approximately below the inferior angle of the scapula, calliper held vertically); and (iv) suprailiac (measured approximately 3 cm above the iliac crest with the calliper held vertically). All measures were performed in triplicate at the non-dominant side, with the patient standing, and recorded to the nearest 0.1 cm. The sum total of the skinfolds from the four sites was converted to BD with the following equations:

$$\text{BD (men)} = 1.1533 - 0.0643 \times \log(\text{sum of 4 skinfolds})$$

$$\text{BD (women)} = 1.1369 - 0.0598 \times \log(\text{sum of 4 skinfolds})$$

The BD values derived from skinfolds were then converted to BF% with the formula of Siri[12] as follows:

$$\text{BF\%} = ((4.95/\text{BD}) - 4.5) \times 100$$

Bioelectrical impedance analysis

BIA analysis were performed using a Body Impedance Analyser Akern (Florence, Italy). The study used the software BodyGram Pro (Akern Florence, Italy) and the BIA equations for men and women for TBW and BF% supplied by the manufacturer and derived from Chumlea *et al.*[13] as follows:

$$\text{TBW (men)} = 1.203 + (0.176 \times \text{W}) + \left(0.449 \times \text{H}^2/\text{R}\right)$$

$$\text{TBW (women)} = 3.747 + (0.113 \times \text{W}) + \left(0.45 \times \text{H}^2/\text{R}\right)$$

Where W is the weight, H the height and R the resistance.

Three-compartment model

The 3-C model divides the body into fat, water and mineral, which has been studied as an alternative method to assess body composition compared with data obtained using the 4-C model.[14] BF% for the 3-C model was obtained using the following equation developed by Siri[15]: BF% = 100 (2.11/BD) − (0.78 × w − 1.354), where w corresponds to TBW/weight (both expressed in kg). BD was assessed using the equation developed by Durnin *et al.*,[16] based on four skinfold measurements and TBW was measured using BIA.[17]

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Table 1 Anthropometric and body composition characteristics of the study population at baseline

Characteristics	Females (n = 11)	Males (n = 10)	Total (n = 21)
Crohn's disease, *n*	8	8	16
Ulcerative colitis, *n*	3	2	5
Age, years, mean ± SD	14.3 ± 1.5	14.3 ± 1.06	14.3 ± 1.3
Weight, kg, mean ± SD	49.2 ± 8.6	48.1 ± 10.5	48.7 ± 9.3
Height, cm, mean ± SD	158.9 ± 7.2	158.1 ± 8.7	158.5 ± 7.7
BMI, kg/m^2, mean ± SD	19.4 ± 2.7	19.3 ± 4.0	19.3 ± 3.3
SF BF%, mean ± SD	31.0 ± 3.5	24.3 ± 8–3	27.8 ± 7.0
BIA BF%, mean ± SD	28.2 ± 5.6	25.3 ± 10.2	26.8 ± 8.0
3-C model BF%, mean ± SD	30.6 ± 4.8	25.2 ± 9.4	28.0 ± 7.7
DEXA BF%, mean ± SD	28.2 ± 4.9	24.7 ± 9.4	26.5 ± 7.4

3-C model, three-compartment model; BIA, bioelectrical impedance analysis; DEXA, dual energy X-ray absorptiometry; SD, standard deviation; SF, skinfold technique.

Table 2 Results of agreement of body composition assessed by skinfold thickness equations, bioelectrical impedance analysis, three-compartment (3-C) model and dual energy X-ray absorptiometry (DEXA) at 0, 6 and 12 months

	BF%	Spearman's correlation		Bland–Altman LOA	
	Mean ± SD	ρ value	*P* value	Bias b/w DEXA and method, mean ± SD	95% LOA
All patients					
DEXA	25.7 ± 7.3				
SF	28.0 ± 6.0	0.83	<0.001	2.3 ± 4.3	−6.1 to 10.7
BIA	25.8 ± 7.9	0.91	<0.001	0.06 ± 3.3	−6.4 to 6.5
3-C model	27.8 ± 7.3	0.90	<0.001	2.0 ± 3.3	−4.5 to 8.6
Males					
DEXA	24.1 ± 8.7				
SF	25.4 ± 7.0	0.88	<0.001	1.4 ± 4.7	−7.8 to 10.5
BIA	23.7 ± 8.9	0.87	<0.001	−0.4 ± 3.7	−7.7 to 6.9
3-C model	25.2 ± 8.2	0.88	<0.001	1.2 ± 3.8	−6.3 to 8.6
Females					
DEXA	27.2 ± 5.2				
SF	30.5 ± 3.5	0.64	<0.001	3.2 ± 3.8	−4.2 to 10.7
BIA	27.8 ± 6.3	0.86	<0.001	0.5 ± 2.9	−5.1 to 6.2
3-C model	30.1 ± 5.4	0.82	<0.001	2.9 ± 2.7	−2.3 to 8.2

b/w, between; BIA, bioelectrical impedance analysis; LOA, limits of agreement; SD, standard deviation; SF, skinfold technique.

Dual energy X-ray absorptiometry

DEXA is considered as the gold standard for body composition and bone mineral content assessment. Whole body DEXA-scans were performed by trained technicians using the Discovery A System (Hologic, Hologic Inc., Marlborough, MA, USA). Each patient was lying supine in light clothing without metal, for 5–10 min. The entire body was scanned, X-rays crossing the body in antero-posterior direction, with radiation dose <0.01 mSv.

Statistical analysis

Statistical analysis was performed using Stata version 13.0 (Stata Corp, College Station, TX, USA) and GraphPad Prism version 8.0.0 for Windows, GraphPad Software (San Diego, CA, USA), www.graphpad.com. Means and standard deviations were determined for BF% via the SF technique, BIA, 3-C model and DEXA. Spearman's rank correlation coefficients (ρ) were calculated to assess the dependence between variables generated from the first three methods, as well as with DEXA parameters here used as a reference method. The method of Bland–Altman was used to identify the 95% limits of agreement (LOA) and bias for each prediction method. Significance for the correlation between the x- and y-axes of the Bland–Altman plots were determined at 95% confidence interval ($P < 0.05$).

Results

Subjects' characteristics

Twenty-one paediatric patients with IBD were included (11 girls and 10 boys, mean age 14.3 years, range 12–16 years). Sixteen

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patients had CD and five UC. Mean BMI did not differ between male and female patients. Their characteristics, anthropometric and body composition data at baseline are presented in Table 1 and detailed in the study published by Callias *et al.*[3] Regarding body composition, there is no significant difference between the measurements made at baseline and those including the baseline and analyses at 6 and 12 months. These data can be found in Tables 1 and 2.

Assessment of body composition methods

Body composition results for the reference method DEXA are detailed in Table 2. Spearman correlation coefficients between fat mass derived from skinfold Durnin equation, BIA, 3-C model and fat mass assessed by DEXA ranged between 0.83 (SF) and 0.91 (BIA), all $P < 0.001$. The results of agreement analysis derived from skinfold equation, BIA, 3-C model and fat mass assessed by DEXA are shown in Table 2 and Figure 1. Average difference between BF% derived from BIA showed the best Bland–Altman LOA. It is well known that women have higher BF% than men and therefore SF formulas are adapted according to gender, particularly in the Durnin equation.[11] Table 2 shows that by separating the analyses according to gender, BF% is higher for women than for men but with no impact on the results on Spearman's correlation and Bland–Altman LOA.

Discussion

This research carried out that a three-compartment model is for the moment not superior to other techniques such as BIA, contrary to our initial hypothesis. Adequate nutritional and body composition assessment remains a challenge for the management of paediatric patients with IBD, since most commonly employed methodologies have major constraints: requirement for stillness, use of radiation or inability to provide reliable data. This study consisted of a comparison of different techniques to measure body composition in children with IBD, using DEXA as a reference method. Although several studies have shown that the measurement of body composition using air displacement plethysmography, such as BodPod, is reliable, the paucity of studies in paediatrics and particularly in IBD had motivated us not to make a comparison with this technique.[18,19] This manuscript complements the one reported by Callias *et al.* in which the authors reported a good correlation but poor agreement of fat mass calculated from SF equations from DEXA.[3] Despite the required exposure to level of radiation with multiple scans, DEXA remains the best technique for assessing body composition in the paediatric population, when the 4-C model is not available, and followed by SF approach.[20] In the present study, we compared SF equation, BIA and 3-C model to DEXA, which was used as the gold standard for measurement of BF%. In children with IBD, these methods showed reliable measurements with small differences for BF% and good LOA. Notably, BIA (LOA −6.4 to 6.5) showed the strongest correlation and agreement in comparison to the SF equation (LOA −6.1 to 10.7) and the 3-C model (−4.5 to 8.6) estimating BF% by DEXA. Traditional methods of BF% assessment, such as SF and BIA, rely on an assumption of a constant hydration of the FFM (FFM of 73.2%).[21] Siri's 3-C model has emerged as preferred criterion method of body composition assessment and was developed after identifying the variation in hydration of FFM as the greatest source of error in the latter methods.[15] Our study contrasts with those revealing that multicomponent models have proven to increase the accuracy of BF% analyses compared to SF and BIA alone.[17,22] Since BF% value from the 3-C model included measurements from SF

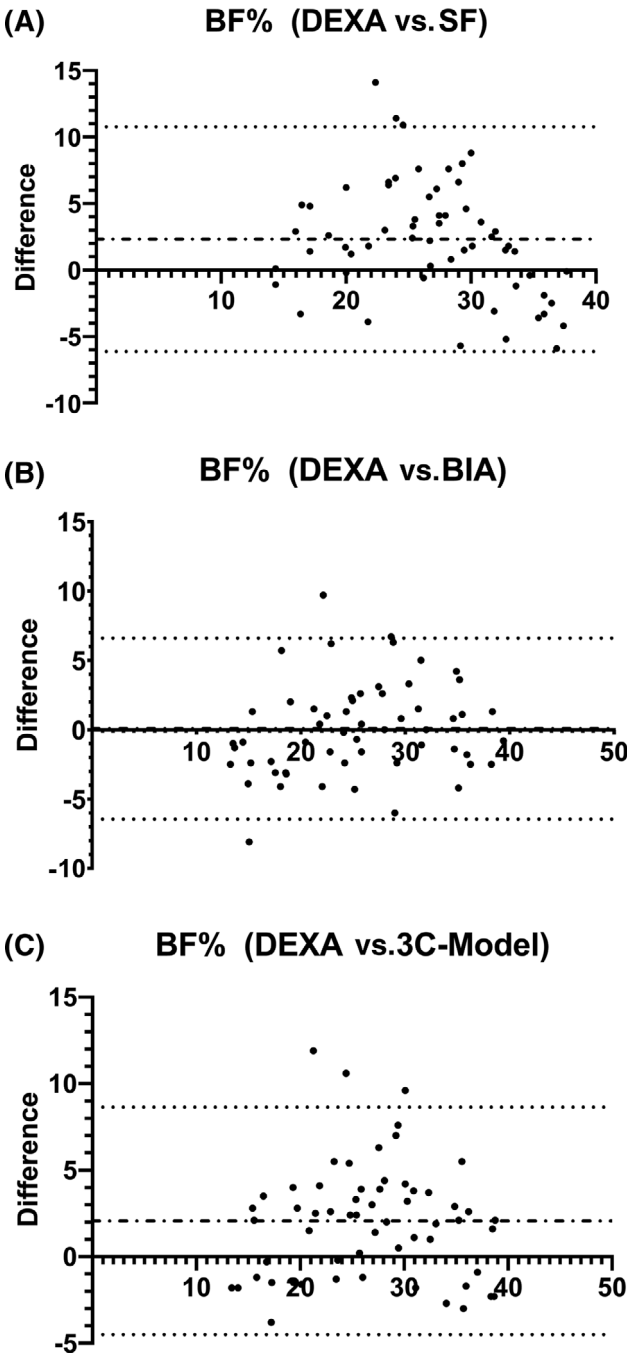

Fig 1 Comparison of predicted BF% between skinfold thickness equations, bioelectrical impedance analysis, three-compartment (3-C) model and DEXA. BIA, bioelectrical impedance analysis; DEXA, dual energy X-ray absorptiometry; SF, skinfold technique.

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and BIA, there could potentially be more bias with the use of a compartment model. A study conducted in a population of young Caucasian male confirmed the limitation of these multi-compartment models related to the estimation of BF% value by using alternative methods such as hydrostatic weighting.[23] The literature also suggests the use of correct prediction equations influenced by age, gender and ethnicity of the source population in order to gain accuracy in the estimates and narrow down the inter-subject's variability in body composition methods.[3] Our study is composed exclusively of a small population of adolescents with IBD and there is no study to date comparing specific equations for this study group. Therefore, the comparison of body composition using the 3-C model versus DEXA is relative and limited due to the use of predictive equations. Overall, our results indicate that body composition assessment using 3-C model is not yet suitable in daily clinical practice in paediatric patients with IBD, and further studies are required before generalising the approach. These studies would be needed to improve the accuracy of this method either by creating new equations more suited for this specific population or by identifying corrective factors, which could be applied to the existing equations. Since the 3-C model has not demonstrated superiority over other methods, potential use of BIA in research and clinical settings of future nutritional intervention and disease status in children with IBD should be considered.

All anthropometric measurements have been performed under the same conditions by the same person, in order to minimise technical errors. Furthermore, our data have been compared to DEXA, the reference method for assessing body composition in children and adults, with high agreement (99%) and reproducibility.[24] Measurements were taken at baseline, 6 and 12 months. This study has several limitations. The study participants were primarily Caucasian, which may limit the generalizability of the findings, as well as the small sample size of patients, rendering the analysis of subgroups difficult. In addition, the 3-C model was constructed with values from the respective methods evaluated, which may increase bias. However, the comprehensive approach of the 3-C model in combining multiple compartments would decrease the potential bias compared with an approach based on only two compartments where more assumptions are made to estimate fat mass of FFM.

Conclusion

The assessment of body composition using bioelectrical impedance analysis provides a valid and accurate method in children with inflammatory bowel disease as compared to dual energy X-ray absorptiometry. In the future, superiority of 3-compartment model in research and clinical settings of nutritional intervention and disease status in children with inflammatory bowel disease remains to be demonstrated.

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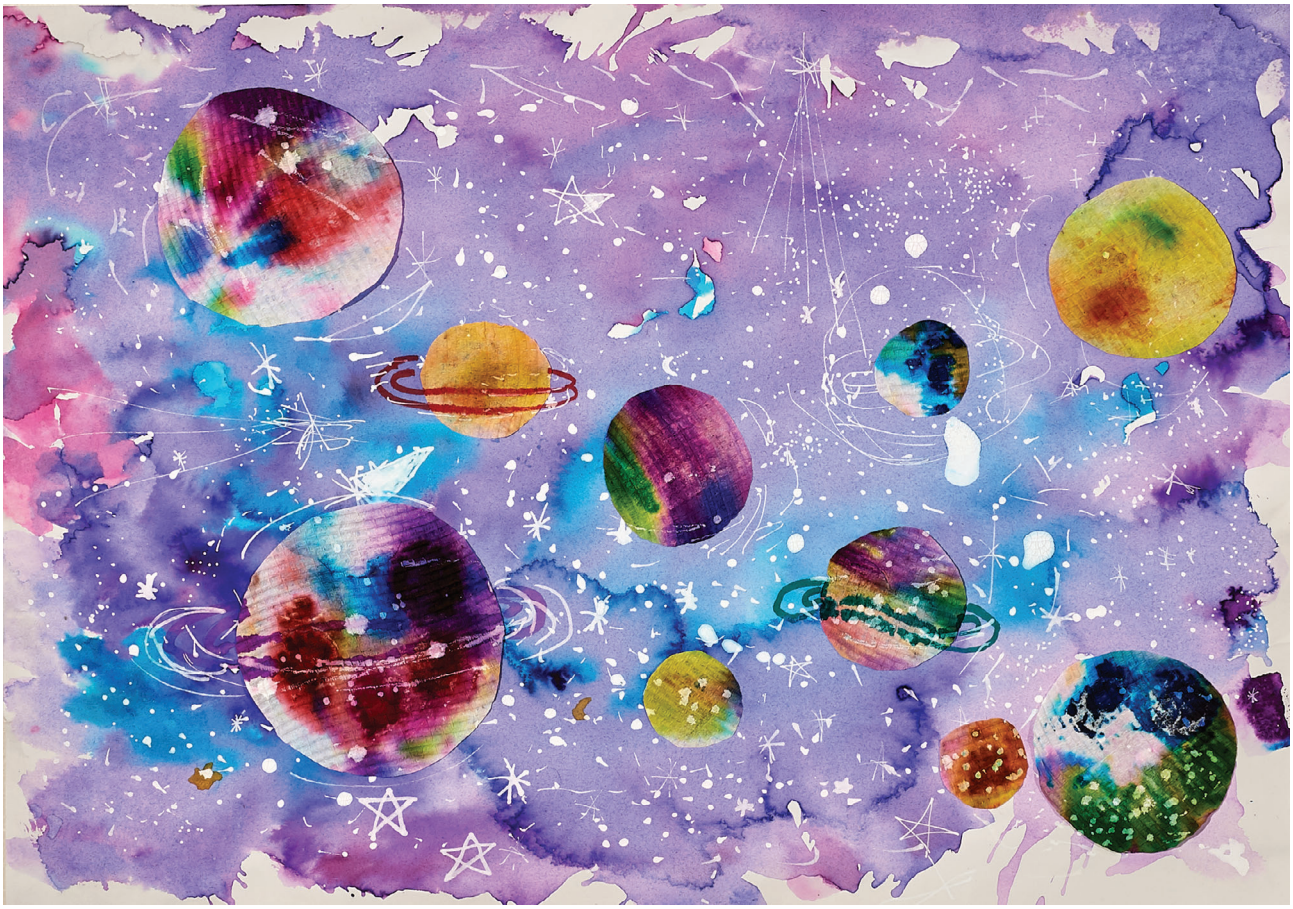

Galaxy by Nina Johnston (7) from WOW Art Competition

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